

STUDENT PERFORMANCE PREDICTION SYSTEM

Using Machine Learning with Streamlit Deployment

Abstract

This project aims to predict student academic performance using Machine Learning algorithms. The system analyzes various factors such as study hours, attendance, sleep hours, parental education, and extracurricular activities to predict exam scores. The best-performing model is deployed using Streamlit.

Introduction

Education plays a crucial role in personal and societal growth. Predicting student performance can help institutions identify weak students early and provide necessary guidance using data-driven methods.

Problem Statement

Educational institutions lack automated systems to predict student performance. Manual evaluation is time-consuming and inefficient. This project builds a predictive ML system.

Objectives

1. Analyze student dataset
2. Perform data preprocessing
3. Train multiple ML models
4. Evaluate model performance
5. Deploy the best model using Streamlit

Dataset Description

The dataset includes features such as Study Hours, Attendance, Sleep Hours, Parental Education, Internet Access, Extracurricular Activities, Previous Scores, and Final Exam Score (Target Variable).

Methodology

- Step 1: Data Collection
- Step 2: Data Cleaning
- Step 3: Exploratory Data Analysis
- Step 4: Feature Engineering
- Step 5: Model Training
- Step 6: Model Evaluation
- Step 7: Deployment

Model Building

Models used:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Random Forest achieved the highest accuracy.

Model Evaluation

Evaluation Metrics Used:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

Random Forest gave the best performance among all models.

Model Deployment

The trained model was saved using Joblib and deployed using Streamlit. The web application allows users to input student details and get predicted exam scores instantly.

Advantages

- Accurate prediction
- Easy to use interface
- Helps educators in decision making
- Scalable and efficient system

Limitations

- Depends on dataset quality
- Limited dataset size
- Cannot capture psychological factors

Conclusion

This project demonstrates how Machine Learning can be effectively used in the education domain to predict student performance and support data-driven academic decisions.

